

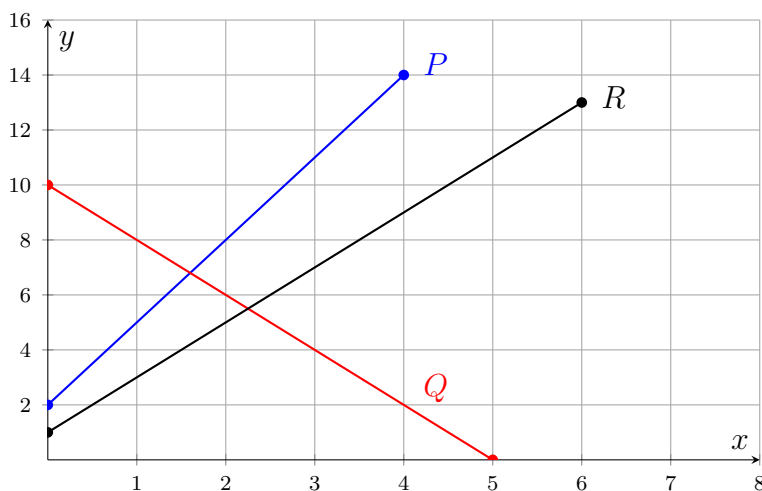
Comparing Rates of Change

Rates of change from tables, from graphs, from equations, and side by side

Directions

- A rate of change is always (change in output) $\div$ (change in input), whatever form the relationship is given in.
- From a table, subtract two rows. From a graph, use two marked points. From an equation $y = mx + b$, the rate is the number multiplying the input.
- When a problem asks you to compare two rates, write the comparison with $>$, $<$, or $=$, keeping the two rates in the order the question names them.

Graph Bank



Lines P, Q and R. Only the problems that name a line by letter use this graph; every other problem on this sheet supplies its own numbers.

1. A gym's total cost after x months is shown in the table.

Months	0	2	4	6
Total cost (\$)	30	54	78	102

- (a) By how much does the total cost change from month 0 to month 6?

Answer: _____

- (b) Find the rate of change, in dollars per month.

Answer: _____

2. A second gym charges $C = 9x + 45$ dollars for x months of membership. What is its rate of change, in dollars per month?

Answer: _____

3. Line P in the Graph Bank passes through the marked points $(0, 2)$ and $(4, 14)$. Find its rate of change.

Answer: _____

4. Ana's running distance is shown in the table. Ben runs according to $d = 160t$ metres after t minutes.

Minutes	0	9
Ana's metres	0	1350

- (a) Find Ana's rate of change, in metres per minute.

Answer: _____

- (b) Compare Ana's rate with Ben's rate using $>$, $<$, or $=$.

Answer: _____

5. A phone battery holds $B = 45 - 6t$ percent of its charge after t hours of use. What is its rate of change, in percent per hour?

Answer: _____

6. Line Q in the Graph Bank passes through the marked points $(0, 10)$ and $(5, 0)$. Find its rate of change.

Answer: _____

7. Line R in the Graph Bank passes through the marked points $(0, 1)$ and $(6, 13)$. A different line has equation $y = 3x$.

(a) Find the rate of change of line R .

Answer: _____

(b) Compare the rate of line R with the rate of the second line, using $>$, $<$, or $=$.

Answer: _____

8. A reading log records the pages left in a book after x days.

Days	0	3	6	9
Pages left	200	155	110	65

(a) By how much does the number of pages left change from day 0 to day 9?

Answer: _____

(b) Find the rate of change, in pages per day.

Answer: _____

9. Machine A makes 3 items per minute; its output is line P in the Graph Bank. Machine B's output is in the table.

Minutes	0	8
Items made by B	5	45

(a) Find machine B's rate of change, in items per minute.

Answer: _____

(b) Compare machine B's rate with machine A's rate using $>$, $<$, or $=$.

Answer: _____

10. A sapling's height is $y = \frac{3}{4}x + 2$ metres after x months. What is its rate of change, in metres per month?

Answer: _____

11. Pool 1 is filling as shown in the table. Pool 2 fills according to $L = 14t + 20$ litres after t minutes.

Minutes	0	6
Litres in pool 1	20	92

(a) Find pool 1's rate of change, in litres per minute.

Answer: _____

(b) Compare pool 1's rate with pool 2's rate using $>$, $<$, or $=$.

Answer: _____

(c) Explain how a graph of the two pools would show which one fills faster.

Answer: _____

12. Delivery service X is described by the table; service Y is described by $d = 60t$ kilometres after t hours.

Hours	0	5
Kilometres for X	0	275

- (a) Find service X's rate of change, in kilometres per hour.

Answer: _____

- (b) Compare service X's rate with service Y's rate using $>$, $<$, or $=$.

Answer: _____

- (c) Explain how you could settle the same question from a graph of the two services without computing either rate.

Answer: _____